BT-202 (GS)

B.Tech., I & II Semester Examination, June 2026

Grading System (GS)

MATHEMATICS-II

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Partial — Q1–Q7 mile (Q8 source me capture nahi hua)

1. (a) Solve: $(y^2 + xy)\,dx + x^2\,dy = 0$. **7**

(b) Solve: $dy/dx + y \tan x = y^2 \sec x$. **7**

2. (a) Solve: $dy/dx - y \sin 2x = \cot x$. **7**

(b) Solve: $(D^2 + 2)\,y = e^x \cos x$. **7**

3. (a) Apply the method of variation of parameters to solve: $d^2y/dx^2 + y = \operatorname{cosec} x$. **7**

(b) Prove that $J_{1/2}(x) = \sqrt{(2/\pi x)} \cdot \sin x$. **7**

4. (a) Solve: $(x + 1)^2\,y'' + (x + 1)\,y' + y = 2 \sin(\log(x + 1))$. **7**

(b) Solve by Charpit's method: $px + qy = pq$. **7**

5. (a) Show that $u = 2 \log(x^2 + y^2)$ is harmonic and find its harmonic conjugate. **7**

(b) Determine the value of p so that the function $f(z) = \tfrac{1}{2} \log(x^2 + y^2) + i \tan^{-1}(px/y)$ is an analytic function. **7**

6. (a) Evaluate $\oint \log z / (z - 1)\,dz$ where C : $|z - 1| = 1/2$, using Cauchy's integral theorem. **7**

(b) Find poles and residues at each pole of $f(z) = 1 / [(z + 1)(z + 2)]$. **7**

7. (a) If $F = \text{grad}\,(x^3 + y^3 + z^3 - 3xyz)$, then find div(F) and curl(F). **7**

(b) Find the directional derivative of $xyz^2 + xz$ at (1, 1, 1) in the direction of the normal to the surface $3xy^2 + y = z$ at (0, 1, 1). **7**

8. (source me ye question available nahi tha — original paper dekhein) **—**

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